Properties of the algorithm

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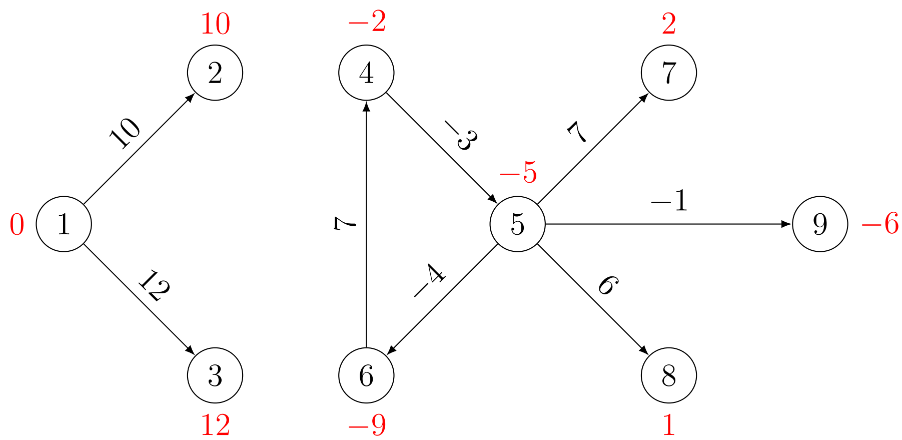
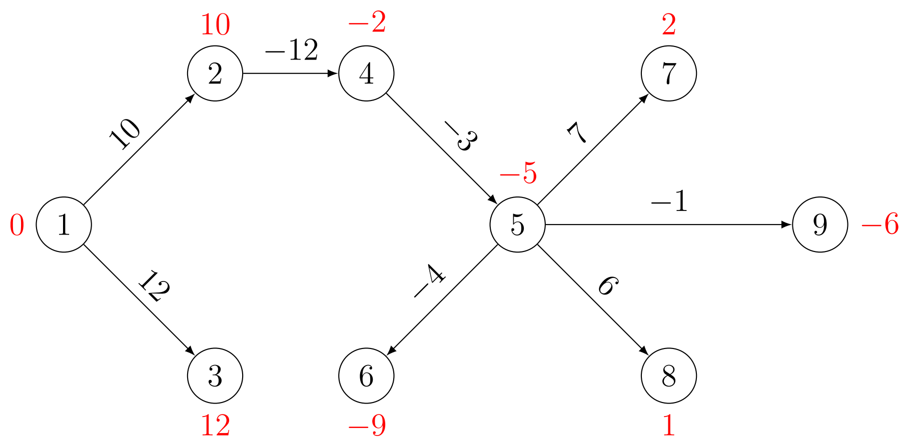
Solution of the practice quiz

To construct Bellman's subnetwork, we need to consider each node separately, and identify one or several arcs that verify the condition with equality:

- Node 2: arc (1,2) verifies $\lambda_2 = \lambda_1 + c_{12} = 10$.
- Node 3: arc (1,3) verifies $\lambda_3 = \lambda_1 + c_{13} = 12$.
- Node 4: two arcs verify the condition:
  - arc (2,4) verifies $\lambda_4 = \lambda_2 + c_{24} = -2$.
  - arc (6,4) verifies $\lambda_4 = \lambda_6 + c_{64} = -2$.
- Node 5: arc (4,5) verifies $\lambda_5 = \lambda_4 + c_{45} = -5$.
- Node 6: arc (5,6) verifies $\lambda_6 = \lambda_5 + c_{56} = -9$.
- Node 7: arc (5,7) verifies $\lambda_7 = \lambda_5 + c_{57} = 2$.
- Node 8: arc (5,8) verifies $\lambda_8 = \lambda_5 + c_{58} = 1$.
- Node 9: arc (5,9) verifies $\lambda_9 = \lambda_5 + c_{95} = -6$.

As node 4 has two arcs verifying Bellman's condition, we can build two Bellman's subnetworks.





We observe that the first Bellman's subnetwork is not a tree: there is a cycle $4 \to 5 \to 6 \to 4$. Note that it is not inconsistent with the result presented in the video, that was based on the assumption that "*each cycle has positive length*". This assumption is not verified here, as the cycle mentioned above has length 0. Therefore, there is no guarantee that Bellman's subnetwork is a tree.

The second subnetwork is indeed a tree. It is a shortest path spanning tree. For instance, the shortest path between node 1 and node 9 is the only path between 1 and 9 in the shortest path spanning tree, that is $1 \to 2 \to 4 \to 5 \to 9$. Its length is

$$10 - 12 - 3 - 1 = -6,$$

which happens to be the optimal label of node 9.